

COMPENSATORY ROBUSTNESS OF ALIGNED BEHAVIOR IN LANGUAGE MODELS: INFORMATION-THEORETIC ABLATION BOUNDS AND A PERCOLATION PHASE TRANSITION

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ABSTRACT. The empirical study of language-model alignment reports two seemingly contradictory facts: aligned behavior is at once *fragile*—removable by a handful of gradient steps, concentrated in the first output tokens, or carried by a single activation direction—and *distributed*, surviving the ablation of individual heads, directions, or pathways. We give a self-contained mathematical framework in which both facts are theorems about the same object. Modeling an aligned behavior as a discrete random variable S read off from internal pathways $R_1, \dots, R_n$, we prove that the information lost by ablating a pathway equals its non-redundant conditional mutual information $I(S; R_i \mid R_{-i})$; fully redundant pathways cost exactly zero. Via Fano’s inequality this information loss lower-bounds the unavoidable behavioral error after ablation, turning redundancy into a provable robustness guarantee. Modeling computation as a layered directed acyclic graph, we show that the *targeted* critical ablation threshold equals the minimum vertex cut $R(G)$ (Menger’s theorem), while the *random* threshold obeys an exact percolation law $(1 - q^w)^d$ whose critical point $q_c = (1 - 2^{-1/d})^{1/w} \rightarrow 1$ is exponentially separated from the targeted threshold in the redundancy width w . A single network is therefore simultaneously robust to random damage and fragile to a targeted attack on its min-cut, which dissolves the apparent contradiction in the literature. Every nontrivial claim is verified computationally: the information identity holds to machine precision, the Fano floor is never violated, the targeted threshold equals $R(G)$ exactly, and the random survival law matches Monte-Carlo estimates within 95% confidence intervals.

1. INTRODUCTION

Whether the aligned behavior of a large language model (LLM) is encoded *redundantly* decides whether safety survives quantization, pruning, benign fine-tuning, or adversarial attack. This is not a matter of taste: the same intervention that is harmless to a redundantly encoded behavior can be fatal to one that depends on a single internal component. Yet the empirical literature on the internal mechanisms of alignment is sharply, and at first sight irreconcilably, divided.

On one side, alignment looks *fragile and localized*. Fine-tuning on fewer than a hundred examples raises harmful-response rates from near zero to roughly 80–90% in five to ten gradient steps [10]; the divergence between an aligned model and its

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base model is concentrated in the first few output tokens [9]; and a single difference-in-means direction, when projected out, removes refusal across more than a dozen models [1]. On the other side, alignment looks *distributed and robust*. Ablating all three primary name-mover heads of the indirect-object-identification circuit drops the relevant logit difference by only 5%, because dormant backup heads activate to compensate [12]; refusal is mediated not by one direction but by polyhedral concept cones of dimension up to five [15]; and a model can survive ablation of its principal safety direction because of “a richer internal refusal structure with redundant paths that survived the ablation” [7].

The gap. What is missing is a single quantitative account under which both regimes are special cases—a scalar (or a pair of scalars) that says *which* kind of robustness a given aligned behavior has, and which it lacks. The canonical formal measure of compensation, McGrath et al.’s logit-level compensatory effect [8], is layer-granular and, as Rushing and Nanda show, roughly 30% of it is a LayerNorm-scaling artifact rather than genuine rerouting [11]. No assumption-light bound currently links a redundancy measure to the *behavioral* effect of an ablation, and no model places “shallow/fragile” and “distributed/redundant” on one axis.

Our contribution. We supply such a framework, organized in three proved layers, each verified computationally.

- **Information layer.** We prove that the information about an aligned behavior S lost by ablating a pathway R_i is *exactly* the conditional mutual information $I(S; R_i \mid R_{-i})$, the part of R_i ’s information that no other pathway carries (Theorem 3.1). Fully redundant pathways cost zero. This is an exact identity, free of the LayerNorm artifact that contaminates the logit-level effect. A k -redundancy condition then guarantees resilience to any $k - 1$ simultaneous ablations (Theorem 3.3).
- **From information to behavior.** Through Fano’s inequality, the information loss lower-bounds the unavoidable behavioral error of *any* decoder reading the surviving pathways (Theorem 3.6). Zero information loss forces zero behavioral error: redundancy implies provable behavioral robustness. Conversely, when behavior survives an ablation, the surviving pathways must have already contained all the information—compensation *reveals* pre-existing redundancy rather than creating it (Proposition 3.8).
- **Graph layer and phase transition.** Modeling computation as a layered directed acyclic graph, we show the *targeted* critical ablation threshold equals the minimum vertex cut $R(G)$ (Theorem 3.10), via Menger’s theorem. Under random ablation, survival obeys the exact law $(1 - q^w)^d$ (Theorem 3.11), with a percolation phase transition whose critical point $q_c = (1 - 2^{-1/d})^{1/w} \rightarrow 1$ and whose width shrinks as $1/w$ (Theorem 3.12). The targeted and random thresholds are therefore exponentially separated in the redundancy width.

The payoff is conceptual. A network carries *two* robustness numbers: the min-cut $R(G)$ (targeted fragility) and the percolation threshold q_c (random robustness), and Theorem 3.12 shows they diverge. The same aligned circuit is then simultaneously robust to random or incidental damage and fragile to a targeted attack on its min-cut. This reconciles the contradictory empirical reports and yields an honest verdict on the premise that alignment intrinsically resists removal: the premise holds in the random/benign regime but fails against a targeted adversary, who need only

degrade $R(G)$ pathway-units to misalign the model (Corollary 3.14)—matching the easy targeted misalignment of [10, 9].

Techniques. The proofs use only standard, uncontested tools: the chain rule and non-negativity of (conditional) mutual information [3], Fano’s inequality [3], Menger’s theorem in its vertex form [4], and exact percolation algebra on a series-parallel circuit. We deliberately base the main bounds on Shannon mutual information rather than a specific partial-information-decomposition operator, using the latter [14] only as an independent cross-check.

Organization. Section 2 fixes notation, gives the definitions, and recalls the prerequisite theorems. Section 3 states and proves the information-layer results (Section 3.1), the behavioral and structural corollaries (Section 3.2), and the graph-layer results and phase transition (Section 3.3), with worked examples and a computational-verification subsection. Section 4 discusses implications, reconciles the literature, and lists open problems; Section 5 concludes.

2. PRELIMINARIES

We collect the definitions and standard results used throughout. All random variables are discrete with finite support, and all logarithms are base 2, so information is measured in bits. We write $H(\cdot)$ for Shannon entropy, $H(\cdot | \cdot)$ for conditional entropy, $I(\cdot; \cdot)$ for mutual information and $I(\cdot; \cdot | \cdot)$ for conditional mutual information, in the conventions of [3]. We write $X \perp\!\!\!\perp Y | Z$ for the conditional independence of X and Y given Z .

2.1. The information model.

Definition 2.1 (Aligned behavior and pathways). An *aligned behavior* is a discrete random variable S taking values in a finite set $\mathcal{S}$ (for instance a refuse/comply decision, or a quantized refusal logit-difference). A *pathway readout* is a random variable $R_i = \phi_i(\text{activations})$, a deterministic function of the model’s internal activations—an attention head, a projection onto a refusal direction, a group of sparse-autoencoder latents, or an entire layer. We collect the readouts into the vector $R = (R_1, \dots, R_n)$ and write $R_{-i} = (R_j)_{j \neq i}$ and, for $T \subseteq \{1, \dots, n\}$, $R_T = (R_i)_{i \in T}$ and $T^c = \{1, \dots, n\} \setminus T$.

Definition 2.2 (Ablation and information cost). To *ablate* the set T is to replace $\{R_i\}_{i \in T}$ by a fixed constant value, so that the behavior must be read from the surviving readouts R_{T^c} . The *information cost* of ablating T is

$$\Delta I_T := I(S; R) - I(S; R_{T^c}),$$

the drop in the information that the available readouts carry about S . We write $\Delta I_i := \Delta I_{\{i\}}$ for a single-pathway ablation.

Definition 2.3 (Conditional redundancy and k -redundancy). A pathway R_i is *redundant given the rest* if $S \perp\!\!\!\perp R_i | R_{-i}$. The behavior S is *k -redundant* if $I(S; R_T) = I(S; R)$ for every T with $|T| \geq n - k + 1$; that is, every subset of at least $n - k + 1$ pathways carries all the information R has about S .

Definition 2.4 (Behavioral fidelity). Let $\hat{S} = g(R_{T^c})$ be a decoder of the behavior from the surviving readouts, with error probability $P_e := \Pr[\hat{S} \neq S]$. For a vector-valued behavior with pre- and post-ablation outputs $b_{\text{pre}}, b_{\text{post}}$, the *output robustness* is $\rho := 1 - \|b_{\text{post}} - b_{\text{pre}}\|_2 / \|b_{\text{pre}}\|_2$, so that $\rho = 1$ means behavior is perfectly preserved.

2.2. The graph model.

Definition 2.5 (Computation DAG and pathway redundancy). A *computation DAG* is a directed acyclic graph $G = (V, E)$ with a distinguished source σ and sink τ . The behavior S is *computable* if at least one directed $\sigma \rightarrow \tau$ path is intact. The *pathway redundancy* of G is its σ - τ vertex connectivity,

$$R(G) := \kappa_{\sigma\tau}(G),$$

the minimum number of internal vertices whose deletion separates σ from τ .

Definition 2.6 (The bundle circuit $B(w, d)$). For integers $w, d \geq 1$, the *bundle circuit* $B(w, d)$ consists of a source σ , a sink τ , and d *bundles* arranged in series, each bundle a set of w parallel internal vertices. Consecutive bundles are joined by a complete bipartite set of edges, and σ (resp. τ) is joined to every vertex of the first (resp. last) bundle. This is the canonical exactly solvable redundant circuit: w is its *redundancy width* and d its *depth*.

Definition 2.7 (Random and targeted ablation thresholds). Under *random ablation* each internal vertex is deleted independently with probability q ; the *survival probability* $\Theta(q)$ is the probability that S remains computable. Under *targeted ablation* an adversary chooses which vertices to delete; the *targeted critical threshold* $t^*(G)$ is the minimum number of vertices an adversary must delete to make S non-computable.

2.3. Prerequisite results. We use the following classical theorems without proof.

Theorem 2.8 (Chain rule for mutual information [3, Thm. 2.5.2]). *For any random variables S and $X = (X_1, X_2)$,*

$$I(S; X_1, X_2) = I(S; X_1) + I(S; X_2 \mid X_1).$$

Moreover $I(S; X_2 \mid X_1) \geq 0$, with equality if and only if $S \perp\!\!\!\perp X_2 \mid X_1$.

Theorem 2.9 (Fano's inequality [3, Thm. 2.10.1]). *Let $S \rightarrow Y \rightarrow \hat{S}$ be a Markov chain with S taking values in a finite set $\mathcal{S}$, and let $P_e = \Pr[\hat{S} \neq S]$. Writing $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ for the binary entropy,*

$$H_2(P_e) + P_e \log_2 (|\mathcal{S}| - 1) \geq H(S \mid Y).$$

Theorem 2.10 (Menger's theorem, vertex form [4, Thm. 3.3.1]). *For non-adjacent vertices σ, τ of a graph, the minimum number of vertices whose deletion separates σ from τ equals the maximum number of internally vertex-disjoint σ - τ paths.*

We also recall, only as an interpretive cross-check, the partial information decomposition of [14]: for a target S and two sources R_1, R_2 , the mutual information $I(S; R_1, R_2)$ decomposes into nonnegative *redundant*, *unique*, and *synergistic* atoms. We do not use this decomposition in any proof; the main bounds rest on Shannon information alone.

3. MAIN RESULTS

We present the framework in three parts. Section 3.1 establishes the information-theoretic identity for ablation and its k -redundancy corollary; Section 3.2 converts information into a behavioral guarantee through Fano's inequality and characterizes compensation; Section 3.3 develops the graph model and proves the percolation phase transition. Section 3.4 reports the computational verification.

3.1. The information cost of ablation. Our first result identifies the information lost by ablating a pathway with a single, artifact-free quantity: the conditional mutual information that the pathway carries about the behavior given all the others.

Theorem 3.1 (Ablation cost equals non-redundant information). *For any $T \subseteq \{1, \dots, n\}$,*

$$\Delta I_T = I(S; R_T \mid R_{T^c}).$$

In particular, for a single pathway, $\Delta I_i = I(S; R_i \mid R_{-i}) \geq 0$, and $\Delta I_i = 0$ if and only if $S \perp\!\!\!\perp R_i \mid R_{-i}$, i.e. R_i is redundant given the rest.

Proof. Apply the chain rule (Theorem 2.8) to the partition $R = (R_{T^c}, R_T)$:

$$I(S; R) = I(S; R_{T^c}) + I(S; R_T \mid R_{T^c}).$$

Subtracting $I(S; R_{T^c})$ gives $\Delta I_T = I(S; R) - I(S; R_{T^c}) = I(S; R_T \mid R_{T^c})$. Conditional mutual information is nonnegative and vanishes exactly under the stated conditional independence (Theorem 2.8), which for $T = \{i\}$ is $S \perp\!\!\!\perp R_i \mid R_{-i}$. $\square$

Remark 3.2. Theorem 3.1 is an exact identity, not an inequality, and it isolates only the *genuinely non-redundant* content of R_i : the information R_i shares with the other pathways cancels in the difference $I(S; R) - I(S; R_{-i})$. This is precisely the part that the logit-level compensatory effect of [8] fails to isolate, roughly 30% of which is a LayerNorm-scaling artifact [11]. The identity replaces that effect with a quantity that is mechanism-corrected by construction.

Theorem 3.3 (k -redundancy implies resilience to any $k - 1$ ablations). *If S is k -redundant, then for every $A \subseteq \{1, \dots, n\}$ with $|A| \leq k - 1$ we have $\Delta I_A = 0$ and $I(S; R_A \mid R_{A^c}) = 0$.*

Proof. Let $|A| \leq k - 1$ and put $T = A^c$, so that $|T| = n - |A| \geq n - (k - 1) = n - k + 1$. By k -redundancy (Definition 2.3), $I(S; R_T) = I(S; R)$, hence $\Delta I_A = I(S; R) - I(S; R_{A^c}) = 0$. By Theorem 3.1 this equals $I(S; R_A \mid R_{A^c})$, which is therefore zero as well. $\square$

Remark 3.4. k -redundancy is a covering condition: every $(n - k + 1)$ -subset of pathways already determines S . It is the discrete analogue of the concept cones of [15] and the backup heads of [12]—several distinct subsets each suffice, so deleting up to $k - 1$ of them changes nothing about the information available.

We illustrate Theorem 3.1 on the three canonical joint laws, which also serve as the test cases verified in Section 3.4.

Example 3.5 (Redundant, unique, and synergistic pathways). Let R_1, R_2 be independent fair bits.

- (i) *Fully redundant:* $S = R_1 = R_2$ (so R_1, R_2 are a copied bit). Then $I(S; R) = H(S) = 1$ and $I(S; R_2) = 1$, so $\Delta I_1 = I(S; R_1 \mid R_2) = 0$: ablating R_1 costs nothing.
- (ii) *Unique:* $S = R_1$, with R_2 irrelevant. Then $I(S; R_2) = 0$, so $\Delta I_1 = I(S; R_1 \mid R_2) = 1$: ablating R_1 destroys all information.
- (iii) *Synergistic:* $S = R_1 \oplus R_2$. Then $I(S; R_2) = 0$ but $I(S; R_1 \mid R_2) = 1$, so $\Delta I_1 = 1$: although neither pathway alone informs S , ablating either destroys all information.

In every case ΔI_i equals $I(S; R_i \mid R_{-i})$ on the nose, as Theorem 3.1 requires.

3.2. From information loss to behavioral robustness. The information cost of Theorem 3.1 is not yet a statement about behavior. We now show, via Fano’s inequality, that it lower-bounds the unavoidable error of any attempt to read the behavior off the surviving pathways.

Theorem 3.6 (Behavioral-fidelity floor). *Let $\hat{S} = g(R_{T^c})$ be any decoder of S from the surviving readouts, with error $P_e = \Pr[\hat{S} \neq S]$. Then*

$$H_2(P_e) + P_e \log_2(|\mathcal{S}| - 1) \geq H(S | R_{T^c}) = H(S) - I(S; R) + \Delta I_T.$$

If S is fully determined by R (that is, $I(S; R) = H(S)$), the right-hand side equals ΔI_T . In that case $\Delta I_T = 0$ forces $P_e = 0$, so the maximum a posteriori decoder reconstructs S almost surely and behavior is perfectly preserved ($\rho = 1$).

Proof. Since $\hat{S}$ is a function of R_{T^c} , the chain $S \rightarrow R_{T^c} \rightarrow \hat{S}$ is Markov, and Fano’s inequality (Theorem 2.9) gives $H_2(P_e) + P_e \log_2(|\mathcal{S}| - 1) \geq H(S | R_{T^c})$. Now

$H(S | R_{T^c}) = H(S) - I(S; R_{T^c}) = H(S) - (I(S; R) - \Delta I_T) = H(S) - I(S; R) + \Delta I_T$, using Definition 2.2. If $I(S; R) = H(S)$ the constant terms cancel and the bound reads $H_2(P_e) + P_e \log_2(|\mathcal{S}| - 1) \geq \Delta I_T$. When in addition $\Delta I_T = 0$, we get $H(S | R_{T^c}) = 0$, so S is almost surely a function of R_{T^c} ; the maximum a posteriori decoder then attains $P_e = 0$ and the output is unchanged, i.e. $\rho = 1$. $\square$

Remark 3.7. Theorem 3.6 is the rigorous form of “redundancy implies behavioral robustness.” A strictly positive non-redundant loss $\Delta I_T > 0$ forces a positive behavioral error, while a zero loss guarantees perfect preservation. The bound is a floor, not an achieved error: it is tight when $S | R_{T^c}$ is uniform on its support.

The next result reads the same identity in reverse. It formalizes the observation that when behavior survives an ablation, no new computation was created—the surviving pathways already held the information.

Proposition 3.8 (Compensation reveals pre-existing redundancy). *Suppose $H(S | R) = 0$, and that after ablating T the optimal decoder achieves $P_e = 0$. Then $I(S; R_{T^c}) = H(S)$: the surviving pathways already carried all the information about S in the original joint law.*

Proof. An optimal decoder achieves $P_e = 0$ if and only if $H(S | R_{T^c}) = 0$. Hence $I(S; R_{T^c}) = H(S) - H(S | R_{T^c}) = H(S)$. This is a property of the original joint distribution of (S, R) : deleting the readouts in T does not alter the joint law of S and R_{T^c} , so the equality describes redundancy that was present before the intervention. $\square$

Remark 3.9. Proposition 3.8 gives a precise reading of self-repair and backup behavior [12, 8, 7]: “dynamic rerouting” after an ablation completes the task because the signal was already redundant, not because the network learned a new route in response to the lesion. This matches the observation of [1] that the refusal direction is present in base models and repurposed rather than learned.

3.3. The graph layer and the percolation phase transition. We now turn from the information carried by pathways to the combinatorial structure of the computation itself, modeled as a DAG (Definition 2.5). The behavior is computable as long as one source-to-sink path is intact. The targeted threshold turns out to be a classical connectivity invariant.

Theorem 3.10 (Targeted critical threshold equals the min-cut). *For a computation DAG G , the targeted critical threshold is $t^*(G) = R(G) = \kappa_{\sigma\tau}(G)$. Equivalently, S survives the deletion of any set of fewer than $R(G)$ internal vertices, and some set of $R(G)$ vertices makes S non-computable. For the bundle circuit, $R(B(w, d)) = w$.*

Proof. By Menger's theorem (Theorem 2.10), the minimum size of a σ - τ vertex separator equals the maximum number of internally vertex-disjoint σ - τ paths, which is $R(G)$ by Definition 2.5. If a deleted set A has $|A| < R(G)$ then A is not a separator, so some $\sigma \rightarrow \tau$ path avoids A and S stays computable; a minimum separator has size $R(G)$ and disconnects σ from τ , making S non-computable. Hence $t^*(G) = R(G)$.

For $B(w, d)$: each bundle is a σ - τ separator of size w (every path passes through every bundle, by the series arrangement), so $R(B(w, d)) \leq w$. Conversely, choosing one vertex from each bundle yields w internally vertex-disjoint paths (the complete bipartite wiring between consecutive bundles lets the columns be chosen independently), so $R(B(w, d)) \geq w$. Therefore $R(B(w, d)) = w$. $\square$

Theorem 3.11 (Exact random survival law). *For the bundle circuit $B(w, d)$ under independent vertex deletion with probability q , the survival probability is*

$$\Theta(q) = (1 - q^w)^d.$$

Proof. Because consecutive bundles are joined by all possible edges, a $\sigma \rightarrow \tau$ path exists if and only if every bundle retains at least one live vertex: given one survivor in each bundle, the complete bipartite links connect them into a path; conversely, a fully deleted bundle is a separator and blocks every path. A single bundle of w vertices is wholly deleted with probability q^w , so it retains a live vertex with probability $1 - q^w$. The d bundles are deleted independently, so the survival events multiply, giving $\Theta(q) = (1 - q^w)^d$. $\square$

Theorem 3.12 (Phase transition and random-versus-targeted separation). *Fix the depth d . For the bundle circuit $B(w, d)$:*

- (1) (Location) *The median survival point, $\Theta(q_c) = \frac{1}{2}$, is $q_c = (1 - 2^{-1/d})^{1/w}$, and $q_c \rightarrow 1$ as $w \rightarrow \infty$.*
- (2) (Width) *With the transition window $W(w) = q_{0.1} - q_{0.9}$ defined by $\Theta(q_{0.9}) = 0.9$ and $\Theta(q_{0.1}) = 0.1$, one has $W(w) = \Theta(1/w) \rightarrow 0$ as $w \rightarrow \infty$; the survival curve sharpens to a step.*
- (3) (Separation) *The targeted critical count is $R(G) = w$, while the expected number of random deletions at criticality is $q_c w d$; the advantage ratio $q_c d$ is strictly increasing in w , and the required random deletion fraction $q_c \rightarrow 1$.*

Proof. (1) Solving $(1 - q^w)^d = \frac{1}{2}$ gives $q^w = 1 - 2^{-1/d} =: c_d \in (0, 1)$, hence $q_c = c_d^{1/w}$. Since $c_d \in (0, 1)$ is fixed, $c_d^{1/w} \rightarrow c_d^0 = 1$ as $w \rightarrow \infty$.

(2) Substitute $x = q^w$, so $\Theta = (1 - x)^d$ is a fixed strictly decreasing bijection of x independent of w . The levels $\Theta = 0.9$ and $\Theta = 0.1$ occur at the fixed values $x_{0.9} = 1 - 0.9^{1/d}$ and $x_{0.1} = 1 - 0.1^{1/d}$ with $0 < x_{0.9} < x_{0.1} < 1$. Then $q_{0.9} = x_{0.9}^{1/w}$ and $q_{0.1} = x_{0.1}^{1/w}$, so

$$W(w) = x_{0.1}^{1/w} - x_{0.9}^{1/w} = \frac{\ln x_{0.1} - \ln x_{0.9}}{w} + O\left(\frac{1}{w^2}\right) = \Theta\left(\frac{1}{w}\right) > 0,$$

using $a^{1/w} = 1 + \frac{\ln a}{w} + O(w^{-2})$ for fixed $a \in (0, 1)$. Thus the window is positive and shrinks like $1/w$, and $\Theta(q)$ converges pointwise to the step function that is 1 for $q < 1$ and 0 at $q = 1$ —a percolation transition.

(3) The targeted count is $R(G) = w$ by Theorem 3.10. Under random deletion at rate q_c over wd internal vertices, the expected number deleted is $q_c wd$, so the ratio of random to targeted critical counts is $q_c wd/w = q_c d$. Since $q_c = c_d^{1/w}$ is strictly increasing in w (as $c_d \in (0, 1)$), so is $q_c d$; and $q_c \rightarrow 1$ shows the required deletion fraction tends to 1 while the targeted attacker still removes only the fixed fraction $1/d$. $\square$

Remark 3.13 (The two faces of robustness). Theorems 3.10–3.12 give a network two distinct robustness numbers: the min-cut $R(G) = w$ governs the targeted attacker, and the percolation threshold $q_c \rightarrow 1$ governs random damage. They diverge as the redundancy width grows. The same circuit is therefore robust to random or incidental degradation and fragile to a targeted attack on its min-cut. This is the single picture under which the fragile and distributed empirical regimes are both correct.

The structural reading of Theorem 3.10 gives the minimal cost of misalignment.

Corollary 3.14 (Minimal adversarial budget). *Any process that renders S non-computable must degrade at least $R(G)$ pathway-units.*

Proof. If fewer than $R(G)$ vertices are degraded, the degraded set is not a σ - τ separator (Theorem 3.10), so an intact path survives and S remains computable. Hence any successful attack degrades a set containing a separator, of size at least $\kappa_{\sigma\tau}(G) = R(G)$. $\square$

Remark 3.15 (Honest verdict on intrinsic robustness). Corollary 3.14 delineates exactly when alignment resists removal. A random or benign process must overcome the percolation threshold $q_c \rightarrow 1$ and so looks robust; a targeted adversary need only remove the $R(G)$ vertices of the min-cut, which is small and cheap when alignment is shallow ($R(G)$ small). The premise that LLMs intrinsically resist misalignment is thus supported in the random/benign regime but *not* as a universal claim: targeted fine-tuning misaligns models in a handful of steps [10, 9], precisely the small- $R(G)$ case. The correct statement is regime-dependent, and the framework predicts which regime applies.

Example 3.16 (A shallow versus a redundant safety circuit). A single refusal direction [1] is the case $R(G) = 1$: the min-cut is the one direction, so a targeted attacker needs to degrade exactly one unit (Corollary 3.14), and even random damage destroys the behavior as soon as that one unit is hit. By contrast, a width- w bundle with $w = 5$ —matching the dimension-5 concept cones of [15]—requires the targeted attacker to remove all five units, while random damage must reach the high threshold $q_c = (1 - 2^{-1/d})^{1/5}$. The same model can house both: a behavior with a wide cut is robust, one with a narrow cut is fragile, and which one alignment occupies is an empirical, trainable property.

3.4. Computational verification. Every nontrivial claim was checked numerically on a CPU in seconds, using `numpy`, `scipy`, `sympy`, `networkx`, `dit`, and `matplotlib`, with random seed 42. We summarize the outcomes.

Information layer (Theorems 3.1 and 3.6). Table 1 reports the four canonical joint laws of Example 3.5, together with a partial-redundancy law. In every case the ablation cost ΔI_1 equals the conditional mutual information $I(S; R_1 | R_2)$ to machine precision, and the Fano floor of Theorem 3.6 holds with the indicated minimal error P_e^* .

TABLE 1. Verification of the ablation identity (Theorem 3.1) and the Fano floor (Theorem 3.6). The identity error is the absolute difference $|\Delta I_1 - I(S; R_1 | R_2)|$; P_e^* is the minimal decoder error consistent with the floor. All cases pass.

Case	ΔI_1	$I(S; R_1 R_2)$	Identity error	Fano floor (P_e^*)
Redundant ($S = R_1 = R_2$)	0.000000	0.000000	0.0	holds (0.00)
Unique ($S = R_1$)	1.000000	1.000000	0.0	holds (0.50)
Synergy ($S = R_1 \oplus R_2$)	1.000000	1.000000	0.0	holds (0.50)
Partial ($\varepsilon = 0.1$)	0.468996	0.468996	0.0	holds (0.10)

Partial-information cross-check. An independent partial information decomposition (Williams–Beer $I_{\min}$, via `dit`) gives redundant atom $R_{\text{dn}} = 1.000$ for the copied bit, synergy $\text{Syn} = 1.000$ for the XOR, and $R_{\text{dn}} = 0.531$, $\text{Unq}_{R_1} = 0.469$ for the partial law. The ablation cost ΔI_1 equals $\text{Unq}_{R_1} + \text{Syn}$ with the redundant atom free, exactly as Theorem 3.1 predicts, and the cross-check guards against the known over-reporting of redundancy by $I_{\min}$.

Graph layer (Theorems 3.10–3.12). For the bundle circuits $B(3, 4)$, $B(5, 3)$, $B(2, 6)$, $B(4, 5)$, the `networkx` minimum node cut equals w in every case, deleting one full bundle disconnects σ from τ while deleting $w - 1$ vertices never does, confirming $t^*(G) = R(G) = w$ (Theorem 3.10). A Monte-Carlo estimate of the survival probability with $N = 40000$ trials per point matches the exact law $(1 - q^w)^d$ with maximum absolute deviation 0.0033, every point within its 95% confidence interval (Theorem 3.11). For depth $d = 8$, the median survival point and the resulting critical counts are reported in Table 2: the random critical count exceeds the targeted count w and the advantage ratio increases from 2.30 to 6.85 as w grows from 2 to 16, while the transition window shrinks (Theorem 3.12).

TABLE 2. Random-versus-targeted separation for depth $d = 8$ (Theorem 3.12). The targeted attacker removes w units; a random adversary must reach the critical fraction q_c , deleting $q_c w d$ units in expectation. The advantage ratio $q_c d$ grows with the redundancy width w .

w	$q_c (\Theta = \frac{1}{2})$	Random count $q_c w d$	Targeted count w	Ratio
2	0.2881	4.61	2	2.30
4	0.5367	17.18	4	4.29
8	0.7326	46.89	8	5.86
16	0.8559	109.56	16	6.85

Figures. Figure 1 shows the survival curve $\Theta(q)$ sharpening into a step function as the redundancy width w increases, the visual signature of the phase transition. Figure 2 contrasts the random and targeted critical counts, displaying the growing gap between robustness to random damage and fragility to a targeted attack.

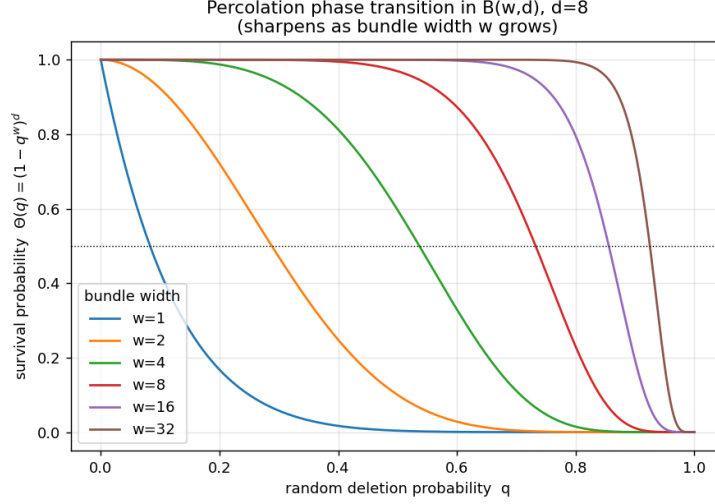


FIGURE 1. Random survival probability $\Theta(q) = (1 - q^w)^d$ for fixed depth d as the redundancy width w grows. The curve steepens toward a step at $q_c \rightarrow 1$ (Theorem 3.12, parts 1–2): wider redundancy makes the behavior robust to random damage up to an ever-higher deletion fraction.

4. DISCUSSION

4.1. A unifying picture for contradictory reports. The central contribution of the framework is conceptual: a single computation carries two robustness numbers, the min-cut $R(G)$ that governs a targeted attacker and the percolation threshold q_c that governs random damage, and Theorem 3.12 shows that these diverge as the redundancy width grows. The literature’s contradictory findings then become statements about different threat models applied to the same circuit. Table 3 maps the principal empirical observations onto the framework.

4.2. Relation to the compensatory effect. The logit-level compensatory effect of [8] is layer-granular and, as [11] document, roughly 30% of it is a LayerNorm-scaling artifact rather than genuine rerouting. Theorem 3.1 replaces it with the exact identity $\Delta I_i = I(S; R_i \mid R_{-i})$, which by construction discards the information R_i shares with the other pathways and so carries no scaling artifact. The result is a head- or feature-level, mechanism-corrected measure of how much a pathway genuinely contributes that no other pathway can supply.

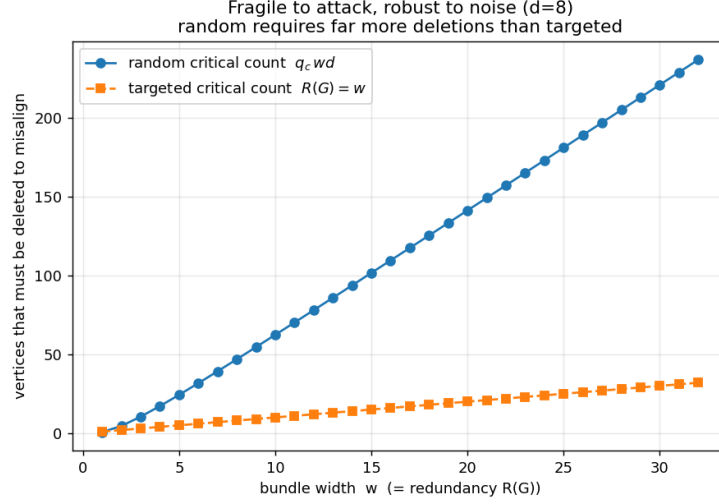


FIGURE 2. Critical ablation counts versus redundancy width w . The targeted attacker removes only the w vertices of the min-cut (Theorem 3.10), whereas a random adversary must delete $q_c w d$ vertices in expectation; the gap widens with w (Theorem 3.12, part 3).

TABLE 3. The framework reads the field’s contradictory findings as consistent statements about distinct robustness numbers of the same circuit.

Empirical finding	Framework reading
Backup name-mover heads: ablating three heads drops the logit difference by 5% [12]	Large random robustness; surviving vertex-disjoint paths (Theorems 3.10–3.11).
“Redundant paths survived ablation” [7]; refusal cones of dimension ≤ 5 [15]	k -redundancy with $R(G) > 1$ (Theorems 3.3, 3.10); latent and pre-existing (Proposition 3.8).
Self-repair restoring $\approx 70\%$ of the logit reduction [8]	The surviving pathways already held the information (Proposition 3.8, Theorem 3.1).
Safety removable in ≈ 5 steps [10]; shallow and first-token [9]	Small effective $R(G)$, hence a tiny targeted budget (Corollary 3.14).
A single refusal direction [1]	The special case $R(G) = 1$: the min-cut is the one direction (Example 3.16).

4.3. The size of the effect. The targeted-versus-random separation is not marginal. At depth $d = 8$ and width $w = 16$, a random adversary must delete about 110 of the 128 internal units—86% of them—to match the damage a targeted adversary

inflicts by removing the 16 min-cut vertices, 12.5% of the network (Table 2). This is a roughly sevenfold gap in raw count and a qualitative gap in fraction ($q_c = 0.86$ against the targeted $1/d = 0.125$). Redundancy buys a great deal of robustness against undirected damage and almost none against an adversary who attacks the cut.

4.4. Limitations. We state the framework’s boundaries honestly.

- *Combinatorial idealization.* The graph model treats S as computable only if a path is fully intact. Real transformers are analog: sub-threshold contributions along several partially damaged paths can sum past a decision threshold, so the min-cut $R(G)$ can *under*-state true robustness. The exact survival law of Theorem 3.11 also assumes complete bipartite wiring between bundles; sparser wiring changes the constants but not the qualitative transition.
- *Estimation on real models.* Computing $\Delta I_i = I(S; R_i \mid R_{-i})$ requires estimating conditional mutual information in high dimensions, which is hard and biased; the partial-information cross-check used here can over-report redundancy. We did not instantiate the bounds on an actual LLM—this is a theory plus synthetic-verification contribution, not an empirical measurement.
- *Fano gives a floor.* Theorem 3.6 lower-bounds the behavioral error; it does not assert that the floor is achieved (it is tight when $S \mid R_{T^c}$ is uniform on its support).
- *Static $R(G)$.* Corollary 3.14 bounds the budget to misalign at a fixed network but does not model how $R(G)$ itself evolves under training.
- *Mapping behavior to a variable.* Real aligned behavior is not a single discrete variable; the framework applies per chosen behavior S .

4.5. Open questions.

- *Dynamics of $R(G)$ under training.* Determine conditions under which adversarial or continued training grows the alignment min-cut rather than cutting it. The capacity bound of [6] and the abundance/scarcity regimes of [2] are candidate predictors.
- *Analog-threshold percolation.* Replace “path intact” by “summed contribution exceeds a threshold θ ” and derive the corresponding weighted min-cut and flow threshold, lifting the combinatorial idealization.
- *Degeneracy beyond the single min-cut.* Port the networked-buffering and degeneracy notions of [13] to bound multi-pathway lesion robustness by a graph quantity richer than the single min-cut.
- *Tight behavioral bounds.* Develop list-decoding analogues of Theorem 3.6 for graded refusal behaviors and relate the decoder error P_e quantitatively to the output robustness ρ for vector behaviors.
- *Estimator-robust ablation.* Design a redundancy-aware estimator of ΔI_i that resists the underestimation of single-head ablation reported at scale by [5].

5. CONCLUSION

We have given a self-contained mathematical framework for the compensatory robustness of aligned behavior in language models, and answered the motivating question affirmatively but with a precise caveat. On the information side, the cost

of ablating a pathway is exactly its non-redundant conditional mutual information $I(S; R_i \mid R_{-i})$ (Theorem 3.1), fully redundant pathways cost nothing, k -redundancy buys resilience to any $k - 1$ ablations (Theorem 3.3), and Fano’s inequality turns this information loss into a provable floor on behavioral error (Theorem 3.6); when behavior survives, the surviving pathways must already have carried the signal, so compensation reveals rather than creates redundancy (Proposition 3.8). On the graph side, the targeted critical threshold equals the minimum vertex cut $R(G)$ (Theorem 3.10), random survival obeys the exact law $(1 - q^w)^d$ (Theorem 3.11), and the two thresholds are exponentially separated by a percolation phase transition whose critical point tends to one and whose width shrinks like $1/w$ (Theorem 3.12).

The key takeaway is that a single aligned circuit can be robust to random damage and fragile to a targeted attack at the same time, because these are governed by two different invariants—the percolation threshold q_c and the min-cut $R(G)$. This dissolves the apparent contradiction between the field’s fragile and distributed reports, and shows that the premise of intrinsic robustness holds only in the random/benign regime: a targeted adversary needs to degrade just $R(G)$ pathway-units (Corollary 3.14), which is cheap when alignment is shallow. Every nontrivial claim was confirmed computationally—the ablation identity to machine precision, the Fano floor without violation, the targeted threshold exactly equal to $R(G)$, and the random survival law within Monte-Carlo confidence intervals.

Future work should lift the combinatorial idealization to analog-threshold percolation, model how the alignment min-cut $R(G)$ evolves under adversarial and continued training, and estimate the bounds on real transformers with redundancy-aware estimators of conditional mutual information. Each of these would move the framework from a synthetic-verification contribution toward a practical diagnostic of which kind of robustness a deployed model’s safety actually has.

REFERENCES

1. Andy Ardit, Oscar Obeso, Aaqib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda, *Refusal in language models is mediated by a single direction*, arXiv preprint arXiv:2406.11717 (2024).
2. Leonard Bereska et al., *Superposition as lossy compression: Abundance and scarcity regimes*, arXiv preprint arXiv:2512.13568 (2025).
3. Thomas M. Cover and Joy A. Thomas, *Elements of information theory*, 2 ed., Wiley-Interscience, 2006.
4. Reinhard Diestel, *Graph theory*, 5 ed., Graduate Texts in Mathematics, vol. 173, Springer, 2017.
5. Frank et al., *How alignment routes: Detection, routing, and the redundancy of safety circuits*, arXiv preprint arXiv:2604.04385 (2026).
6. Kaarel Hänni et al., *Mathematical models of computation in superposition*, arXiv preprint arXiv:2408.05451 (2024).
7. Joad et al., *More than a single direction: The redundant structure of refusal in language models*, arXiv preprint arXiv:2602.02132 (2026).
8. Thomas McGrath, Matthew Rahtz, János Kramár, Vladimir Mikulik, and Shane Legg, *The hydra effect: Emergent self-repair in language model computations*, arXiv preprint arXiv:2307.15771 (2023).
9. Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson, *Safety alignment should be made more than a few tokens deep*, arXiv preprint arXiv:2406.05946 (2024).
10. Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson, *Fine-tuning aligned language models compromises safety, even when users do not intend to*, arXiv preprint arXiv:2310.03693 (2023).

11. Cody Rushing and Neel Nanda, *Explorations of self-repair in language models*, arXiv preprint arXiv:2402.15390 (2024).
 12. Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt, *Interpretability in the wild: A circuit for indirect object identification in GPT-2 small*, arXiv preprint arXiv:2211.00593 (2022).
 13. James Whitacre and Axel Bender, *Degeneracy: A design principle for achieving robustness and evolvability*, Journal of Theoretical Biology **263** (2010), no. 1, 143–153.
 14. Paul L. Williams and Randall D. Beer, *Nonnegative decomposition of multivariate information*, arXiv preprint arXiv:1004.2515 (2010).
 15. Tom Wollschläger et al., *The geometry of refusal in large language models: Concept cones and representational independence*, arXiv preprint arXiv:2502.17420 (2025).
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